MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 12:38 AM, 06 July 2024

Question # 2 of 10 (Start time: 12:40:02 AM, 06 July 2024) Total Marks: 1

The expression $\frac{(n+1)!}{s^{n+2}}$ is obtained by substituting the Laplace transform of t^n in the inductive step(last step). What does this expression represent?

Select the correct option

Reload Math Equations

- Laplace transform of t^{2n}
- Laplace transform of t^{n+2}
- Laplace transform of t^n
- Laplace transform of t^{n+1}

MTH647 - Methods in Mathematical Physics (Quiz - 03) **Quiz Start Time: 12:38 AM, 06 July 2024**

Question # 3 of 10 (Start time: 12:41:29 AM, 06 July 2024) **Total Marks:** 1

What is the base case(1st step) when proving the Laplace transform of t^n using mathematical induction?Where n is whole number.

Select the correct option

Reload Math Equations

- ○ $n=1$
- ◉ $n=0$
- ○ $n=-1$
- ○ $n=2$

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 12:38 AM, 06 July 2024

Question # 4 of 10 (Start time: 12:42:01 AM, 06 July 2024) Total Marks: 1

What is the Laplace transform of $e^{-2t}sint$ using the $s-shifting\ theorem$,

if the Laplace transform of $sin\ (t)$ is $\frac{1}{s^2+1}$?

Select the correct option

Reload Math Equations

- (●) $\frac{1}{s^2+4}$
- ( ) $\frac{1}{(s-2)^2+1}$
- ( ) $\frac{1}{(s+2)^2+1}$
- ( ) $\frac{1}{s^2+1}$

MTH647 - Methods in Mathematical Physics (Quiz - 03) **Quiz Start Time: 12:38 AM, 06 July 2024**

Question # 5 of 10 (Start time: 12:43:39 AM, 06 July 2024) **Total Marks:**

What condition must a function $f(t)$ satisfy for the Laplace transform to exist according to the existence theorem?

Select the correct option

Reload Math Equations

- ○ It must be piecewise differentiable.
- ◉ It must be piecewise continuous and satisfy a boundedness condition.
- ○ It must be periodic with a bounded period.
- ○ It must be continuous everywhere.

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 12:38 AM, 06 July 2024

Question # 6 of 10 (Start time: 12:44:05 AM, 06 July 2024) Total Marks: 1

Which function is defined as $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$?

Select the correct option

Reload Math Equations

- Euler Function
- Gamma Function
- Riemann Function
- Laplace Function

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 12:38 AM, 06 July 2024

Question # 7 of 10 (Start time: 12:45:28 AM, 06 July 2024) Total Marks: 1

What is the Laplace transform of $e^{at}f(t)$ denoted as?

Select the correct option

Reload Math Equations

- $F(s \times a)$
- $F(s - a)$
- $F(s)$
- $F(s + a)$

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 12:38 AM, 06 July 2024

Question # 8 of 10 (Start time: 12:46:24 AM, 06 July 2024) Total Marks: 1

The Laplace transform of $f(t) = t^3$ is given by:

Select the correct option

Reload Math Equations

- $\frac{6}{s^4}$
- $\frac{4}{s^3}$
- $\frac{3}{s^4}$
- $\frac{3s}{s^4}$

MTH647 - Methods in Mathematical Physics (Quiz - 03) **Quiz Start Time: 12:38 AM, 06 July 2024**

Question # 9 of 10 (Start time: 12:47:16 AM, 06 July 2024) **Total Marks:** 1

What is the Laplace transform of the second derivative $f''(t)$ of a function $f(t)$?

Select the correct option

Reload Math Equations

- $s^2F(s)+sf(0)-f'(0)$
- $s^2F(s)-sf(0)+f'(0)$
- $s^2F(s)-sf(0)-f'(0)$ (selected)
- $s^2F(s)+sf(0)+f'(0)$

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 12:38 AM, 06 July 2024

Question # 10 of 10 (Start time: 12:47:55 AM, 06 July 2024) Total Marks: 1

Where can discontinuities occur in a piecewise continuous function?

Select the correct option

Reload Math Equations

- (●) Discontinuities can occur at the endpoints of each interval and within the intervals.
- ( ) Only in the middle of each interval.
- ( ) Only at the endpoints of each interval.
- ( ) Only at the critical points of the function.

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 01:06 AM, 07 July 2024

Question # 2 of 10 (Start time: 01:08:24 AM, 07 July 2024) Total Marks: 1

What is the Laplace transform of $e^{-2t}sint$ using the $s-shifting\ theorem$,

if the Laplace transform of $sin\ (t)$ is $\frac{1}{s^2+1}$?

Select the correct option

Reload Math Equations

- (●) $\frac{1}{(s+2)^2+1}$
- ( ) $\frac{1}{(s-2)^2+1}$
- ( ) $\frac{1}{s^2+1}$
- ( ) $\frac{1}{s^2+4}$

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 01:06 AM, 07 July 2024

Question # 3 of 10 (Start time: 01:09:45 AM, 07 July 2024) Total Marks:

The expression $\frac{(n+1)!}{s^{n+2}}$ is obtained by substituting the Laplace transform of t^n in the inductive step(last step). What does this expression represent?

Select the correct option

Reload Math Equations

- (●) Laplace transform of t^{n+1}
- ( ) Laplace transform of t^{n+2}
- ( ) Laplace transform of t^{2n}
- ( ) Laplace transform of t^n

Question # 4 of 10 (Start time: 01:10:51 AM, 07 July 2024) Total Marks: 1

Where can discontinuities occur in a piecewise continuous function?

Select the correct option

Reload Math Equations

- (●) Discontinuities can occur at the endpoints of each interval and within the intervals.
- ( ) Only at the critical points of the function.
- ( ) Only at the endpoints of each interval.
- ( ) Only in the middle of each interval.

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 01:06 AM, 07 July 2024

Question # 5 of 10 (Start time: 01:11:52 AM, 07 July 2024) Total Marks: 1

The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded by a constant value. What is the significance of this boundedness?

Select the correct option

Reload Math Equations

- It guarantees the continuity of the function.
- It guarantees the differentiability of the function.
- It ensures the periodicity of the function.
- (●) It ensures the existence of the Laplace transform.

MTH647 - Methods in Mathematical Physics (Quiz - 03) **Quiz Start Time: 01:06 AM, 07 July 2024**

Question # 6 of 10 (Start time: 01:12:50 AM, 07 July 2024) **Total Marks:** 1

What condition must a function ***f(t)*** satisfy for the Laplace transform to exist according to the existence theorem?

Select the correct option

Reload Math Equations

- (●) It must be piecewise continuous and satisfy a boundedness condition.
- ( ) It must be piecewise differentiable.
- ( ) It must be periodic with a bounded period.
- ( ) It must be continuous everywhere.

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 01:06 AM, 07 July 2024

Question # 7 of 10 (Start time: 01:13:15 AM, 07 July 2024) Total Marks: 1

What does the $s - shifting\ theorem$ state?

Select the correct option

Reload Math Equations

- The Laplace transform of $e^{-at}f(t)$ is $F(s) \cdot a \cdot$
- The Laplace transform of $e^{-at}f(t)$ is $F(s) - a.$
- The Laplace transform of $e^{-at}f(t)$ is $F(s + a).$
- The Laplace transform $of\ e^{-at}f(t)$ is $F(s - a).$

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 01:06 AM, 07 July 2024

Question # 9 of 10 (Start time: 01:14:38 AM, 07 July 2024) Total Marks: 1

The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded for all $t \geq 0$. What does M represent in this condition?

Select the correct option

Reload Math Equations

- (●) Upper bound on the function
- ( ) Minimum value of the function
- ( ) Maximum value of the function
- ( ) Lower bound on the function

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 01:06 AM, 07 July 2024

Question # 10 of 10 (Start time: 01:15:38 AM, 07 July 2024) Total Marks:

What is the Laplace transform of the derivative of a function $f(t)$ denoted by $f\prime(t)$?

Select the correct option

Reload Math Equations

- $\mathcal{L}\{f(t)\}$
- $s\mathcal{L}\{f(t)\} - f(0)$
- $s\mathcal{L}\{f(t)\}$
- $f(0) - s\mathcal{L}\{f(t)\}$

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 06:11 PM, 07 July 2024

Question # 1 of 10 (Start time: 06:11:57 PM, 07 July 2024) Total Marks: 1

If $f(t) = \sin t$, then the Laplace transform of $f'(t)$ is:

Select the correct option

Reload Math Equations

- $\mathcal{L}(f\prime(t)) = \frac{-s}{s^2+1}$
- $\mathcal{L}(f\prime(t)) = \frac{s^2}{s^2+1}$
- $\mathcal{L}(f\prime(t)) = \frac{s}{s^2+1}$
- $\mathcal{L}(f\prime(t)) = \frac{-s^2}{s^2+1}$

MTH647 - Methods in Mathematical Physics (Quiz - 03) **Quiz Start Time: 06:11 PM, 07 July 2024**

Question # 4 of 10 (Start time: 06:14:01 PM, 07 July 2024) **Total Marks:** 1

What is the Laplace transform of the second derivative $f''(t)$ of a function $f(t)$?

Select the correct option

Reload Math Equations

- (●) $s^2F(s)-sf(0)-f'(0)$
- ( ) $s^2F(s)+sf(0)-f'(0)$
- ( ) $s^2F(s)-sf(0)+f'(0)$
- ( ) $s^2F(s)+sf(0)+f'(0)$

MTH647 - Methods in Mathematical Physics (Quiz - 03) Quiz Start Time: 12:38 AM, 06 July 2024

Question # 1 of 10 (Start time: 12:38:23 AM, 06 July 2024) Total Marks: 1

What is the Laplace transform of t^a expressed in terms of Gamma function?

Select the correct option

Reload Math Equations

- $\frac{(a-1)!}{s^{a+1}}$
- $\frac{(a)!}{s^{a+1}}$
- $\frac{\Gamma(a+1)}{s^{a+1}}$
- $\frac{\Gamma(a+1)}{s^{a}}$